Data Science and Visualization

Unit II

DESCRIPTIVE STATISTICS

KUMARAGURU
college of technology
character is life

KCED
Learning Augmented

Unit II – Descriptive Statistics

Advanced Attribute Types - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis - Visual Outlier Identification - Missing Data Visualization - Exploratory Data Analysis (EDA) Techniques - Histograms, KDEs, Box Plots, and Violin Plots.

CO2	**Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.**

Unit II – Descriptive Statistics

Deep Dive into Data/Attribute Types: Nominal-Ordinal-Binary-Numeric -Discrete vs. Continuous-Asymmetric Attributes-Measurement Scales - Basic Statistical Descriptions: Measures of Central Tendency -Mean-Median-Mode - Measures of Dispersion -Range-Quartiles-Variance-Standard Deviation-IQR - Understanding Skewness and Kurtosis -Conceptual - Data Cleaning Context: Visual identification of outliers - Visualizing Missing Data –Types of Missingness- Exploratory Data Analysis Visualization Techniques-Univariate: Histograms with Kernal Density Estimation-Box Plots -Violin Plots.

CO2	**Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.**

Missing Data or Incomplete Data

- Occurs when no data value is stored for a variable.

- When data points are incomplete, models can become **biased**, results may be **inaccurate**, and performance might **degrade**, leading to **unreliable outcomes.**

V1	V2	V3	V4
X	.	X	X
X	X	X	.
X	X	.	X
X	X	X	.

Missing Data or Incomplete Data

Missing values appear when some entries in a dataset are left blank, marked as NaN, None or special strings like "Unknown".

School ID	Name	Address	City	Subject	Marks	Rank	Grade
101.0	A	123 Main St	Mumbai	Math	85.0	2	B
102.0	B	456 Oak Ave	Delhi	English	92.0	1	A
103.0	C	789 Pine Ln	Bengaluru	Science	78.0	4	C
NaN	D	101 Elm St	Chennai	Math	89.0	3	B
105.0	E	NaN	Kolkata	History	NaN	8	D
106.0	F	222 Maple Rd.	NaN	Math	95.0	1	A
107.0	G	444 Cedar Blvd	Pune	Science	80.0	5	C
108.0	H	555 Birch Dr	Jaipur	English	88.0	3	B

What causes Missing Data?

1. Human Error/Data Entry Issues
2. Sensor Malfunction/System Failures
3. Privacy Concerns/Sensitive Data
4. Conditional Missing Data
5. Survey Design/Sampling Issues

What causes Missing Data?

Missing data can arise from several sources, and knowing why it occurs helps in choosing the right action.

Common causes:

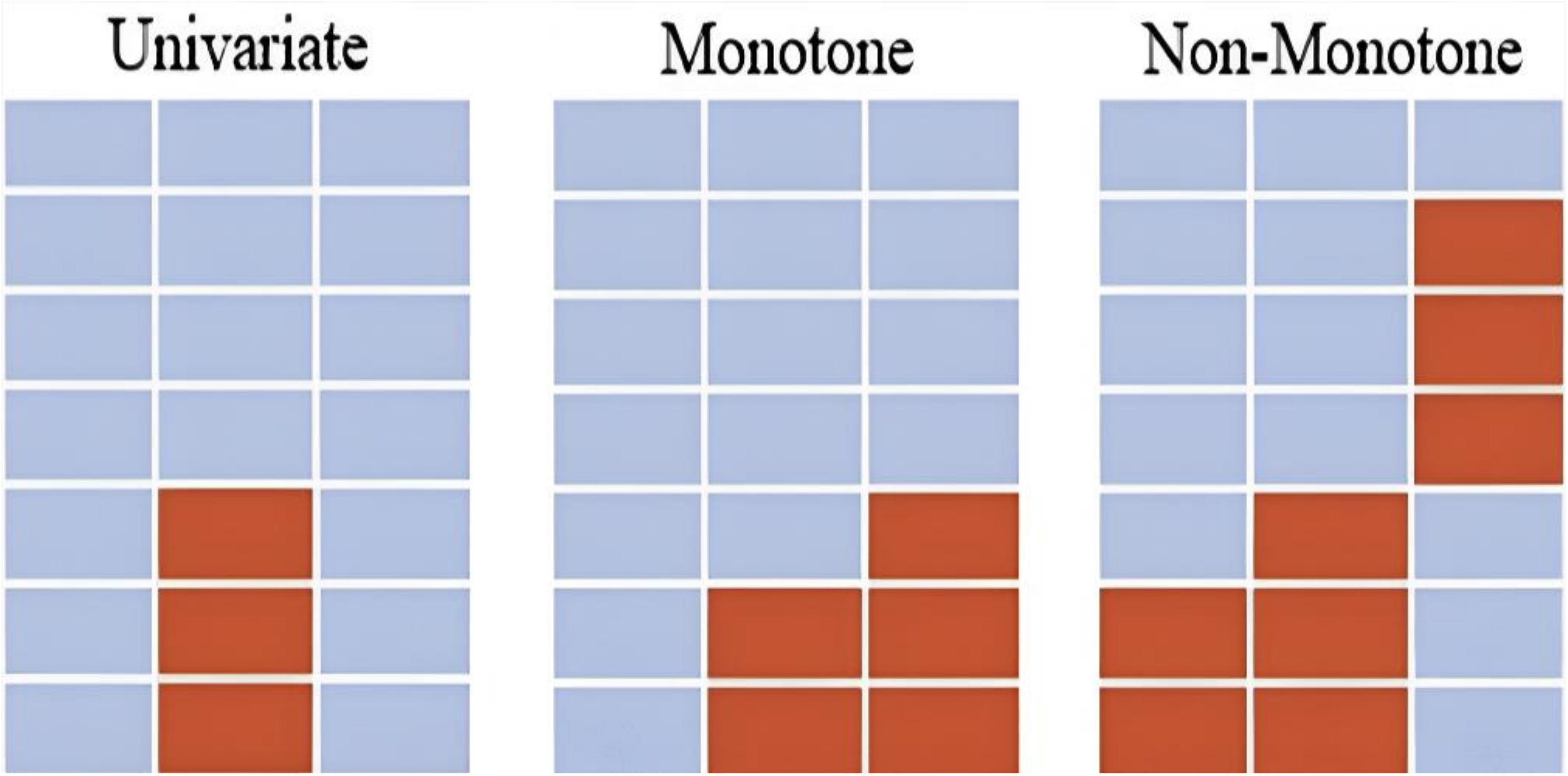
- **Technical issues:** Failures in data collection, sensors, or transmission errors.
- **Human errors:** Incorrect data entry or processing mistakes.
- **Privacy constraints:** Deliberate omission of sensitive information.
- **Data processing problems:** Errors during cleaning or transformation.

Understanding the cause helps assess whether missingness may introduce bias and guides decisions like **imputation or removal.**

Problems of Missing Data

- **Loss of efficiency**
- **Complications** in handling and analysing the data
- **Bias** resulting from differences between missing and complete data
- Inappropriate or **wrong predictions**

Common Patterns of Missing Data



Patterns of Missing Data

- Univariate

  The pattern is univariate when only one variable is missing data.

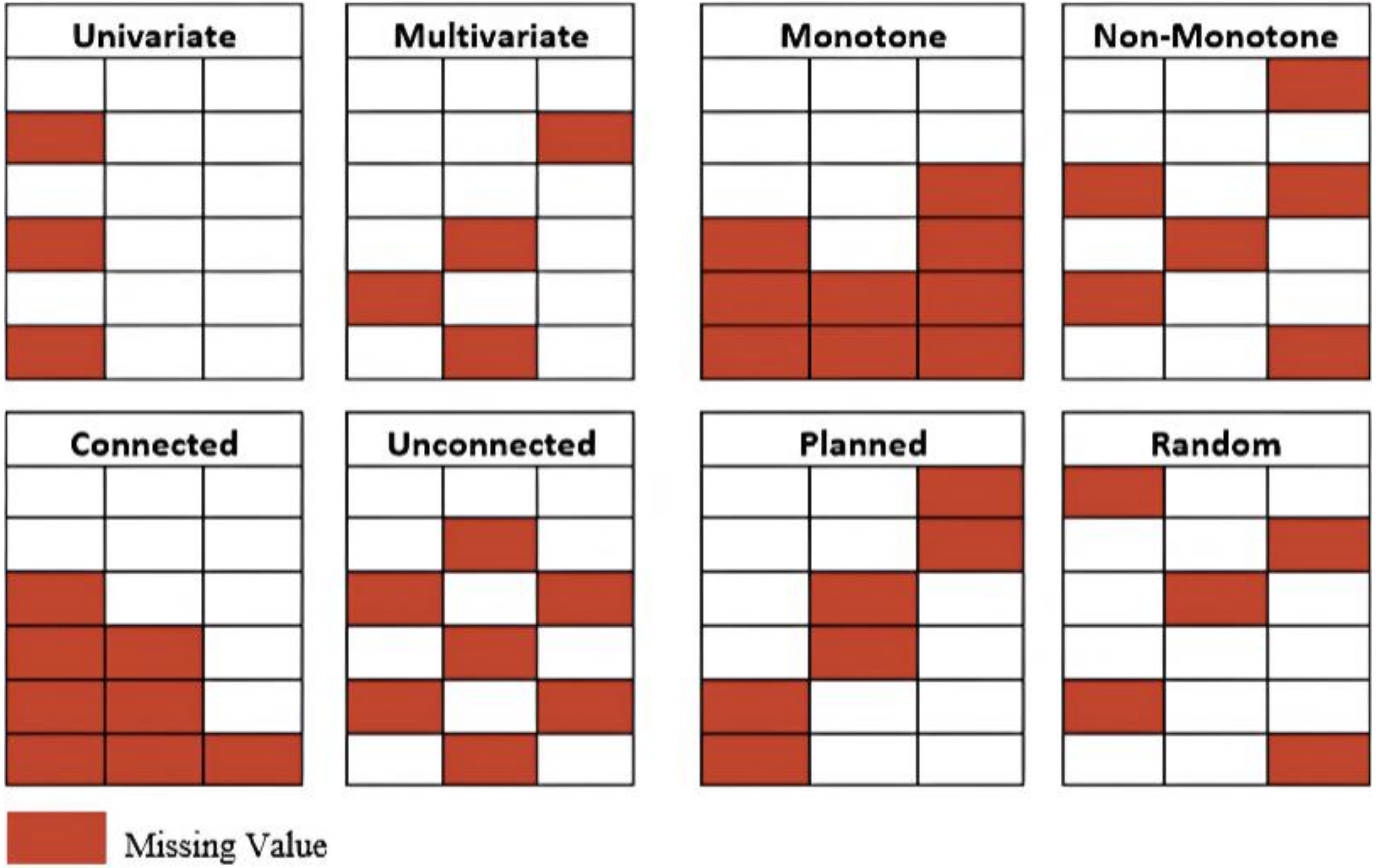
- Monotone

  Variables in the data can be organized in a specific order.

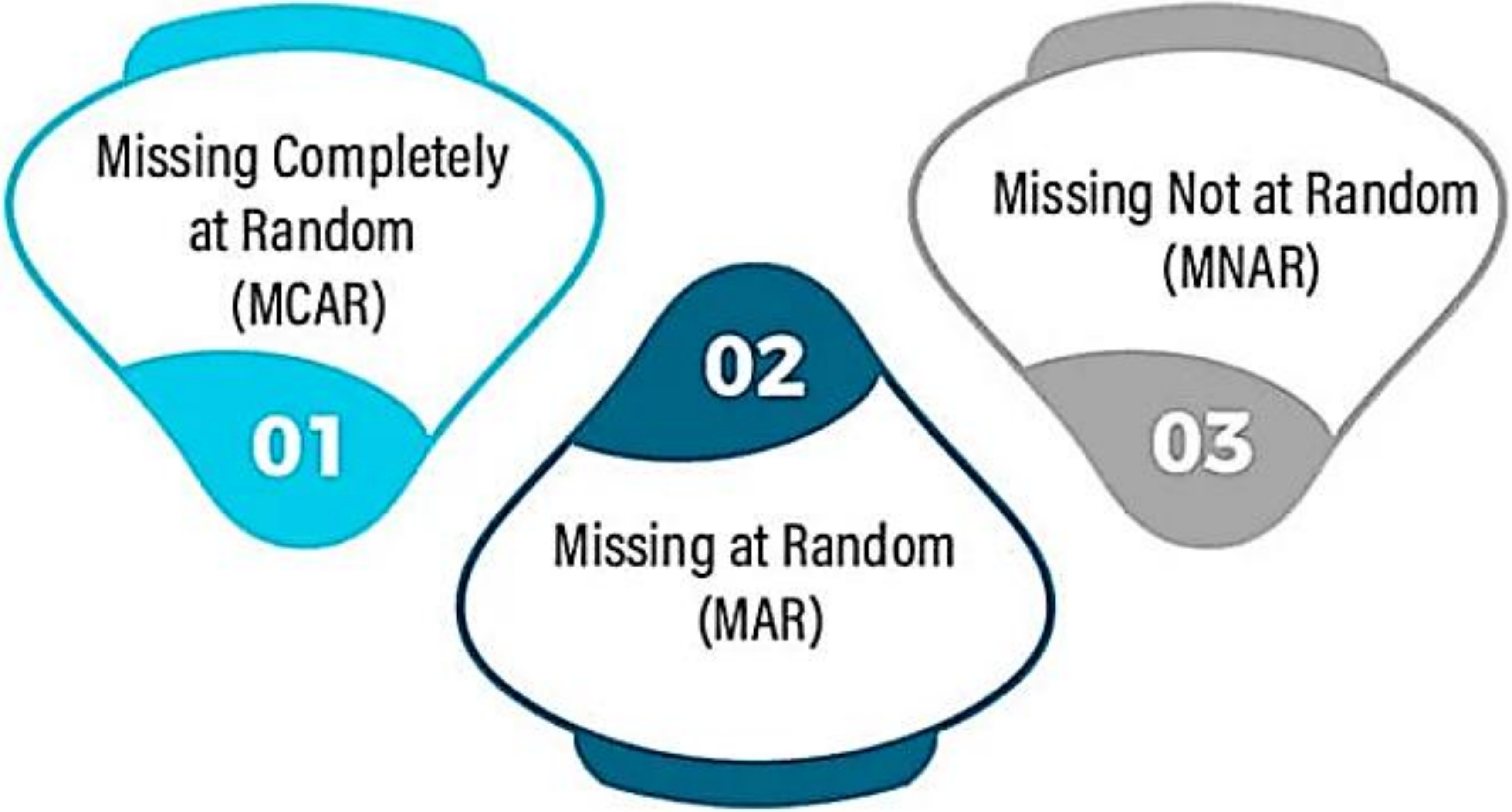
- Non-monotone

  Where the absence of one variable does not influence the absence of any other variables.

Other Patterns of Missing Data



Types of Missing Data



Missing Completely at Random (MCAR)

- The probability of data missing on X is unrelated to the value of X or to values on other variables in the dataset.

- Refers to a situation where the chance of data being missing in a case is unrelated to both the observed and the missing values of that case.

- **Example**: A customer accidentally skips a question in an online form due to distraction, with no connection to the question's content or their previous answers.

Missing at Random (MAR)

- When data is MAR, the missingness can often be predicted using the other observed variables.
- It is possible to estimate the missing values from the observed data, and there is a statistical relationship between the observed data and the missing variables.
- **Example**: For example, in a medical dataset, younger patients might be less likely to report their cholesterol levels than older patients. Here, missingness is related to age but not cholesterol values.

Missing Not at Random (MNAR)

- The missing values are dependent on unobserved information, making it the hardest type to handle.

- Refers to a scenario where neither the Missing Completely at Random (MCAR) nor the Missing at Random (MAR).

- **Example**: In a survey on income, higher-income respondents might be less likely to disclose their income, making it more difficult to infer or predict their actual earning.

Why Handling Missing Data is Crucial

- Biased Results and Loss of Accuracy
- Difficulty in Model Convergence
- Risk of Discarding Valuable Information
- Distortion of Data Distributions
- Increased Model Complexity
- Impact on Real-World Applications

Methods for Identifying Missing Data

Functions	Descriptions
.isnull()	Detect missing values
.notnull()	Detect non-missing values
.info()	Summary with missing counts
.isna()	Same as isnull()
dropna()	Remove missing rows/columns
fillna()	Fill missing values
replace()	Replace specific values
drop_duplicates()	Remove duplicate rows
unique()	Get unique values in a Series/DataFrame

Representation of Missing Values in Datasets

Common representations include:

- **Blank cells:** Empty entries in spreadsheets or CSV files.
- **Special values:** Placeholders like "NA", "NaN", "NULL" or numbers such as -999.
- **Codes or flags:** Labels like "MISSING" or "UNKNOWN" to indicate specific types of missing data.

Missing Data Visualization Techniques

1. Missingness Matrix (Heatmap / Map)

- Displays data as a grid where missing values are highlighted.
- Helps detect patterns (random vs systematic missingness).
- Useful for spotting blocks, trends, or feature-specific gaps.

2. Bar Plot of Missing Values

- Shows count or percentage of missing values per feature.
- Quickly identifies problematic variables.

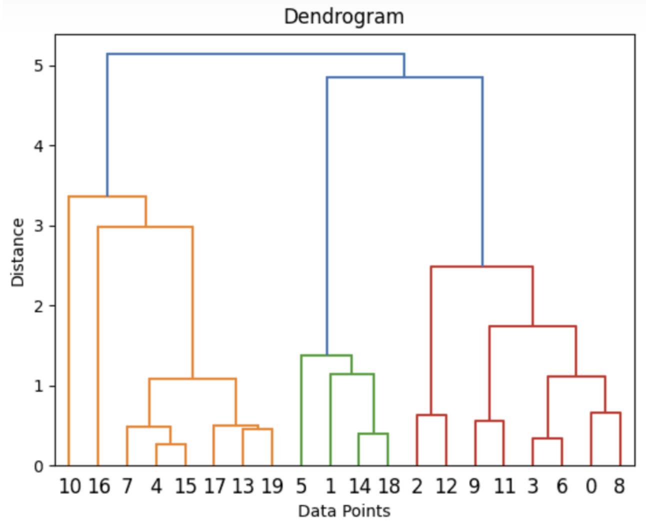
3. Missingness Correlation Heatmap

- Visualizes correlation between missingness of variables.
- Helps detect dependencies (e.g., if one variable missing → another likely missing).

Missing Data Visualization Techniques

4. Dendrogram / Hierarchical Clustering

- Groups variables based on similarity of missingness patterns.
- Useful for diagnosing structural issues.



5. Pattern Plot / Aggregated Missingness Plot

- Shows combinations of missing values across variables.
- Helps identify frequent missing-data patterns.

6. Time-based Missingness Plot (for temporal data)

- Shows missing values over time.
- Detects sensor failures, outages, or seasonal gaps.

7. Pairwise Scatterplots with Missing Indicators

- Reveals relationships between missingness and observed values.
- Helps infer MCAR, MAR, or MNAR behavior.

Missing Data Visualization Techniques

Popular Tools / Libraries:

- Python: missingno, seaborn, matplotlib, pandas
- R: VIM, naniar, Amelia

Interactive Python Notebook File

1. Create Example Dataset with Missing Values

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Reproducibility
np.random.seed(0)

# Create dataset
n = 100
data = pd.DataFrame({
    "Feature_A": np.random.randn(n),
    "Feature_B": np.random.randn(n),
    "Feature_C": np.random.randn(n),
    "Feature_D": np.random.randn(n)
})

# Introduce missing values
data.loc[np.random.choice(n, 20, replace=False), "Feature_A"] = np.nan
data.loc[:30, "Feature_B"] = np.nan
data.loc[data["Feature_C"] > 1, "Feature_C"] = np.nan
data.loc[np.random.choice(n, 10, replace=False), "Feature_D"] = np.nan
```

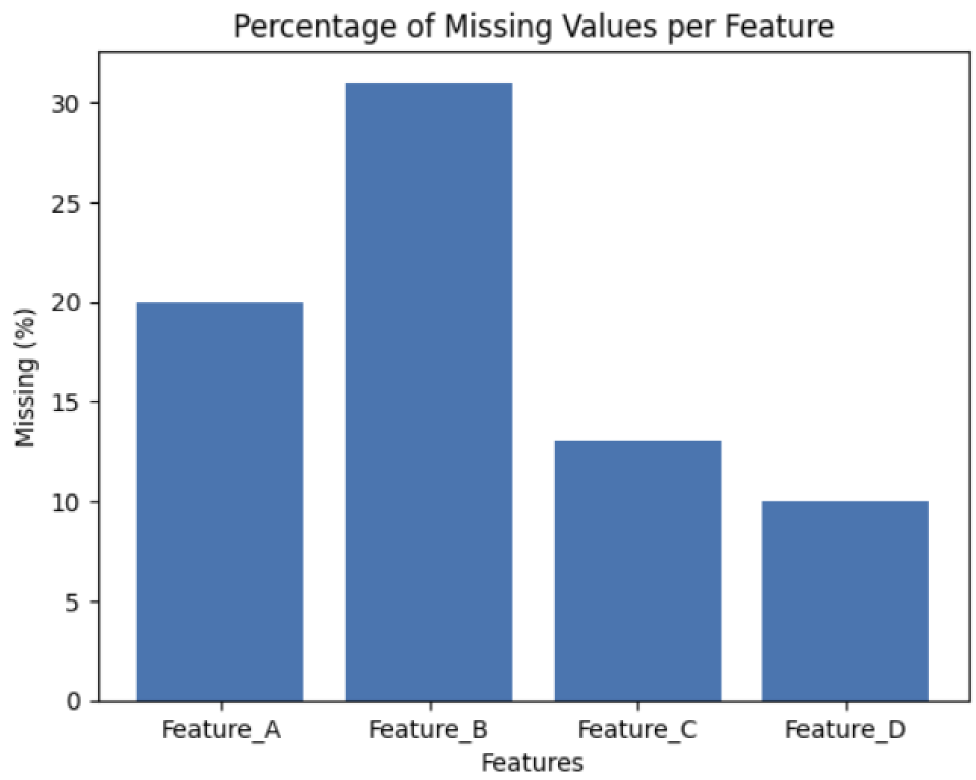
Missing Data Visualization Techniques

1 Missing Values Bar Plot

Quickly shows which features are problematic.

In real research, this helps decide:

- Drop variable?
- Impute?
- Investigate data collection issue?

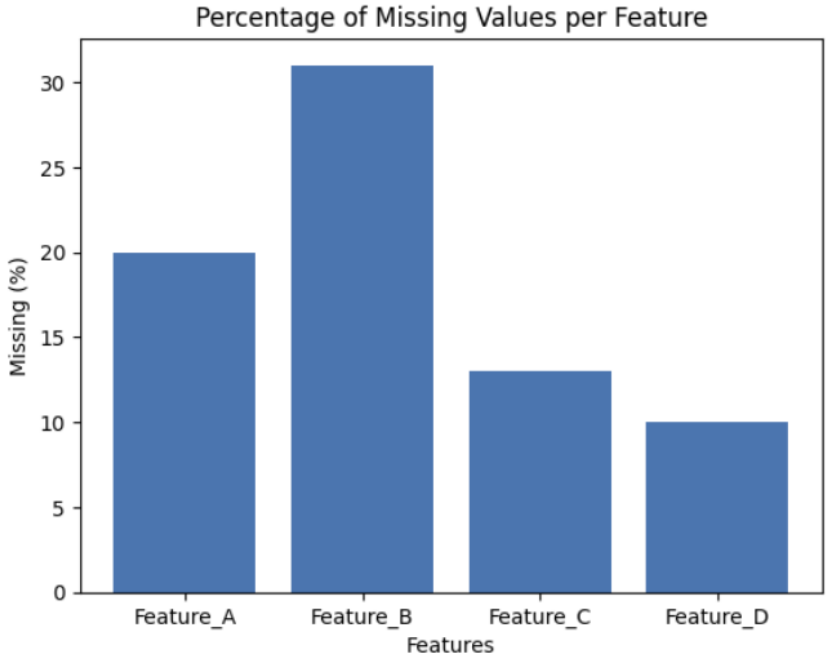


Missing Data Visualization Techniques

✅ Plot 1: Missing Values Bar Plot

```
missing_percent = data.isnull().mean() * 100

plt.figure()
plt.bar(missing_percent.index, missing_percent.values)
plt.title("Percentage of Missing Values per Feature")
plt.xlabel("Features")
plt.ylabel("Missing (%)")
plt.show()
```



✅ **Simple Interpretation**

- Feature_B = 30% missing → Potential problem
- Feature_D = 5% missing → Usually manageable

Missing Data Visualization Techniques

2 Missingness Matrix

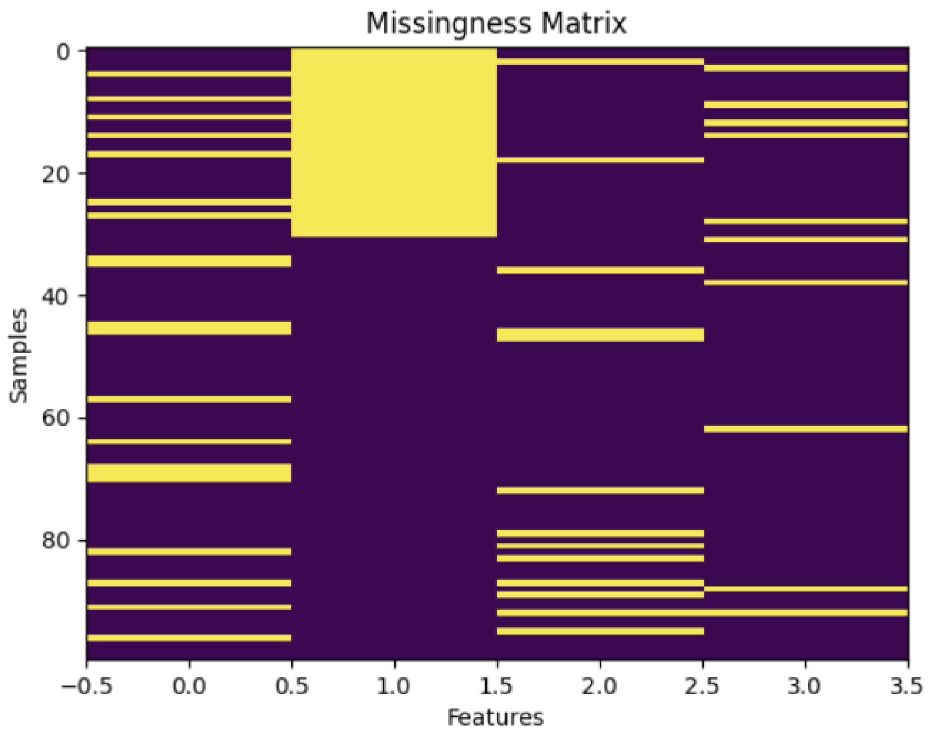
→ Reveals patterns.

You can visually detect:

- Random scatter → likely MCAR
- Blocks → system/sensor failure
- Structured gaps → MAR/MNAR suspicion

A visual map of missing vs available data:

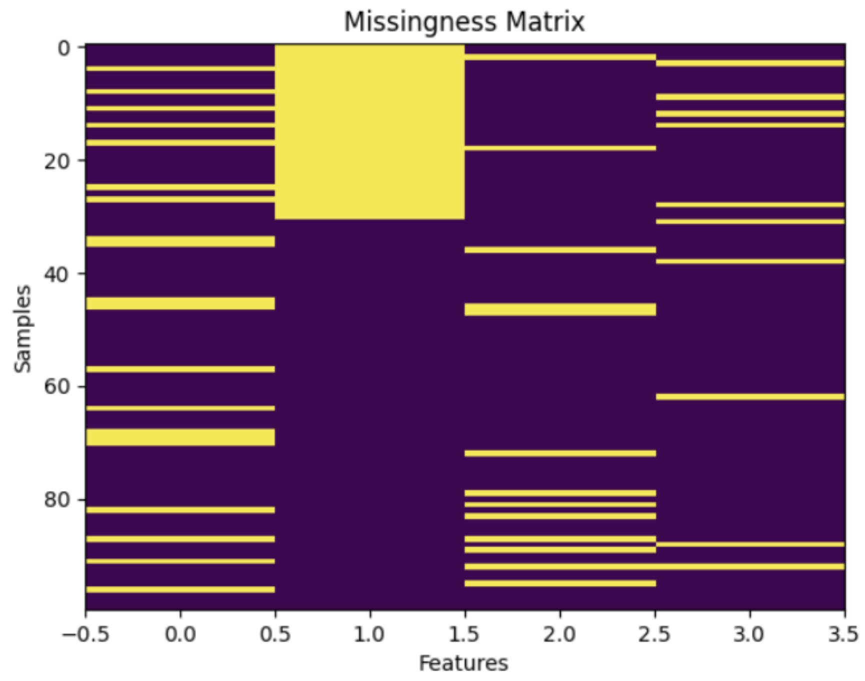
- Colored → missing
- Dark → present



Missing Data Visualization Techniques

✅ Plot 2: Missingness Matrix

```
plt.figure()
plt.imshow(data.isnull(), aspect='auto')
plt.title("Missingness Matrix")
plt.xlabel("Features")
plt.ylabel("Samples")
plt.show()
```



✅ What This Plot Shows

- ✓ Visual map of missing values
- ✓ Each row = record
- ✓ Each column = feature

Missing values appear as highlighted blocks.

✅ Simple Interpretation

Patterns you may see:

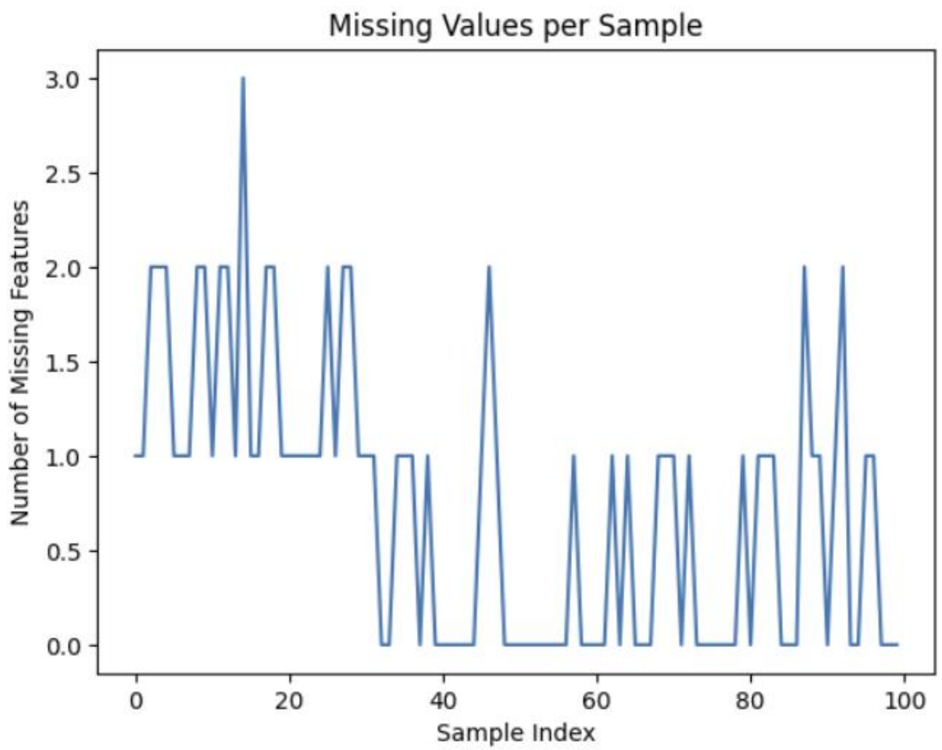
- ✓ Random scattered → MCAR
- ✓ Large blocks → Data collection failure
- ✓ Structured gaps → Systematic missingness

Missing Data Visualization Techniques

3 Missing Values per Sample Plot

Helps detect:

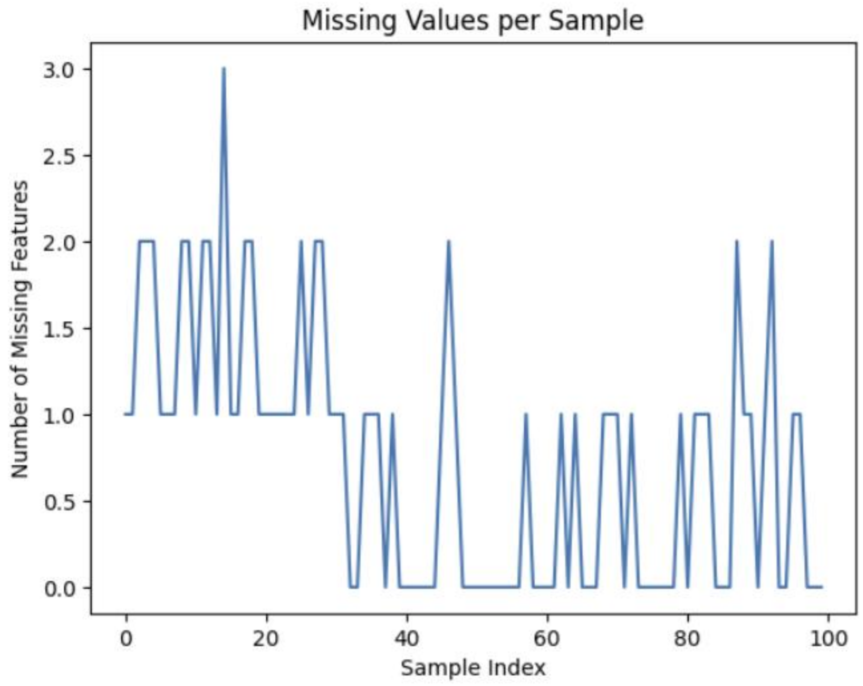
- Outlier records
- Corrupted rows
- Time/system failures (especially in logs / IoT / medical / NLP datasets)



Missing Data Visualization Techniques

✅ Plot 3: Missing Values per Sample

```
plt.figure()
plt.plot(data.index, data.isnull().sum(axis=1))
plt.title("Missing Values per Sample")
plt.xlabel("Sample Index")
plt.ylabel("Number of Missing Features")
plt.show()
```



✅ **Simple Interpretation**

Helps detect:

- ✓ Corrupted records
- ✓ Outlier rows
- ✓ Faulty observations

Missing Data Visualization Techniques

Summary

Plot	Helps With
Bar Plot	Identify problematic variables
Matrix	Detect missingness patterns
Per Sample Plot	Detect bad records

Strategies for Handling Missing Values in Data Analysis

Depending on the nature of the data and the missingness, several strategies can help maintain the integrity of our analysis.

Some of the most effective methods to handle missing values:

- **Removing Rows with Missing Values**
- **Imputation Methods**
  - ✓Mean, Median and Mode Imputation
  - ✓Forward and Backward Fill
- **Interpolation Techniques**
  - ✓Linear Interpolation
  - ✓Quadratic Interpolation

Click Here for Example - Code - Output

Impact of Handling Missing Values

Handling missing values is important to ensure the dataset is reliable and suitable for analysis.

- **Improved data quality:** A dataset with fewer missing values is more complete and dependable for analysis.
- **Better model performance:** Machine learning models work more effectively when trained on complete data, leading to more accurate predictions.
- **Maintaining data consistency:** Properly treating missing values (through imputation or removal) helps keep the dataset structured and consistent.
- **Reduced bias:** Addressing missing values prevents distortion in results, allowing the analysis to better reflect the true patterns in the data.

How To Handle Missing Data in Python

https://github.com/aarthyviven/DataScienceNotebook/blob/main/Missing_Data_Visualization.ipynb

https://github.com/swarnava-96/Exploratory-Data-Analysis-EDA/blob/main/EDA3.ipynb

https://github.com/Sukanyasingh3/Exploratory-data-analysis/tree/main/EDA

References

https://www.analyticsvidhya.com/blog/2021/10/handling-missing-value/

https://www.geeksforgeeks.org/machine-learning/handling-missing-values-machine-learning/

https://www.kaggle.com/code/selahattinsanli/visualizing-missing-data

https://cran.r-project.org/web/packages/naniar/vignettes/naniar-visualisation.html

SINCE 1984

FOUR DECADES OF NATION BUILDING THROUGH EXCELLENCE IN EDUCATION

KUMARAGURU

ENGINEERING | LIBERAL ARTS | AGRICULTURE | MANAGEMENT
BUSINESS INCUBATION | TAMIL RESEARCH

KUMARAGURU COLLEGE OF TECHNOLOGY character is life
KCT BUSINESS SCHOOL
KUMARAGURU COLLEGE OF LIBERAL ARTS AND SCIENCE
KUMARAGURU INSTITUTE OF AGRICULTURE
KUMARAGURU SCHOOL OF BUSINESS
KRiA KUMARAGURU RESEARCH & INNOVATION ALLIANCE
KUMARAGURU SCHOOL OF INNOVATION
KUMARAGURU KC.IRI CENTRE FOR INDUSTRIAL RESEARCH & INNOVATION
FORGE
SEA SAKTHI EXCELLENCE ACADEMY
N.MAHALINGAM TAMIL RESEARCH CENTRE
KCT TECH PARK
KARE Kumaraguru Action For Relief And Empowerment
Dr.N.MAHALINGAM CHESS ACADEMY

THANK YOU